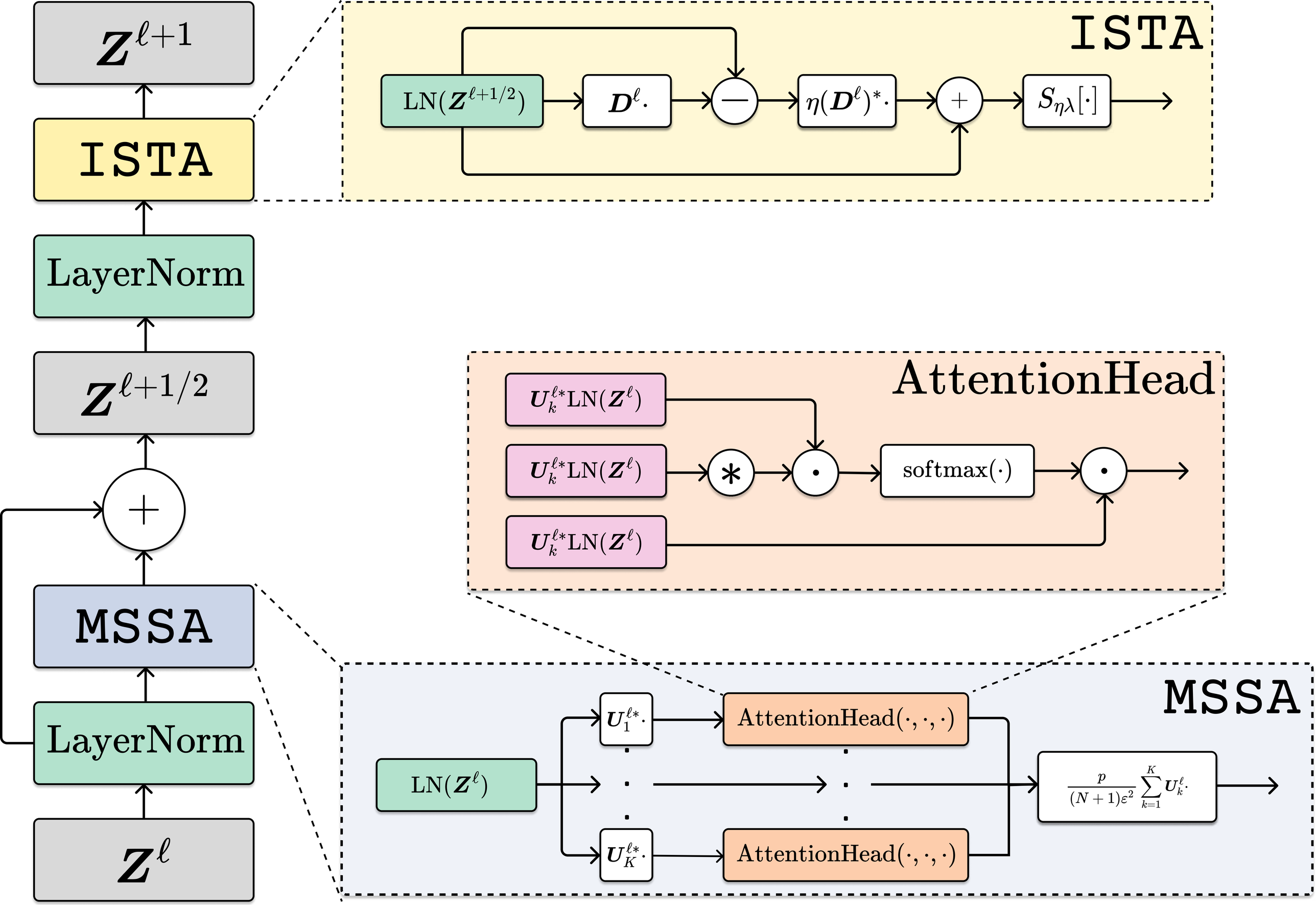
$\boldsymbol{Z}^{\ell+1}$
ISTA
LayerNorm
$\boldsymbol{Z}^{\ell+1/2}$
MSSA
LayerNorm
$\boldsymbol{Z}^{\ell}$
ISTA
$\mathrm{LN}(\boldsymbol{Z}^{\ell+1/2})$
$\boldsymbol{D}^{\ell}\cdot$
$\eta(\boldsymbol{D}^{\ell})^*\cdot$
$S_{\eta\lambda}[\cdot]$
AttentionHead
$\boldsymbol{U}_k^{\ell *}\mathrm{LN}(\boldsymbol{Z}^{\ell})$
$\boldsymbol{U}_k^{\ell *}\mathrm{LN}(\boldsymbol{Z}^{\ell})$
$\boldsymbol{U}_k^{\ell *}\mathrm{LN}(\boldsymbol{Z}^{\ell})$
softmax$(\cdot)$
MSSA
$\mathrm{LN}(\boldsymbol{Z}^{\ell})$
$\boldsymbol{U}_1^{\ell *}\cdot$
$\boldsymbol{U}_K^{\ell *}\cdot$
AttentionHead$(\cdot,\cdot,\cdot)$
AttentionHead$(\cdot,\cdot,\cdot)$
$\frac{p}{(N+1)\varepsilon^2}\sum_{k=1}^{K}\boldsymbol{U}_k^{\ell}\cdot$